

Name:
Z number:

Total Points: 15

COP 3410-001: Data Strc Algrthm Anlys Python

Florida Atlantic University

Recursion Worksheet

NOTE:

- Each question carries 1 point.
- All solutions must be answered in the space labeled “put your solutions here.”
- submit the completed worksheet in the form of a PDF.
- The file name must be *YourName_Recursion Worksheet.pdf*

1. Which data structure is responsible for keeping track of function calls during recursion? choose the correct option. and justify your answer.
 - a. Queue
 - b. Stack
 - c. LinkedList
 - d. Graph

Put Your Option Here:

Justification:

2. What would happen if the base case is not defined in the recursive method? choose the correct options from the below. and justify your answer.
- a. Stack Overflow
 - b. Stack Underflow
 - c. Program Crashes
 - d. Infinite Loop

Put Your option(s) Here:

Justification:

3. what will be the output of the following program? Show recursive trace for the fun(3)?
what will be the output of print(fun(5))?

```
def fun(n):  
    if n == 4:  
        return n  
    else:  
        return 2 * fun(n + 1)  
  
print(fun(3))
```

Put Your Solution Here:

4. Given a sorted list with indices ranging from 0 to 50, and assuming there is a recursive binary search function called `search()`, what would be the sequence of function calls made by `search()` when searching for an item located at index 6? The initial function call is `search(0, 50)`.

choose the correct option. and justify your answer.

- a. `search(0, 25) search(0, 12) search (0, 6)`
- b. `search(0, 25) search(0, 12)`
- c. `search (0, 25) search (0, 12) search (0, 6) search (0, 3)`

Put Your Solution Here:

Justification:

5. Which searching algorithm works best for the following array:
arr = [18,23,11,9,365,11]
choose the correct option. and justify your answer.
- a. Binary Search
 - b. Linear Search

Put Your Solution Here:

Justification:

-
6. What is returned as the output due to the call fun(2,9)?

```
def fun(x, y):  
    if y == x:  
        return x  
    elif y > x:  
        return fun(x, y - x)  
    else:  
        return fun(x - y, y)
```

Put Your Solution Here:

7. Will fun(50) iterate infinitely? What is returned as a result of the call fun(50)?

```
def fun(y):  
    if y == 0:  
        return y-5  
    else:  
        return fun(y-10)
```

Put Your Solution Here:

-
8. What is the output of the following function code?

```
def fun(n):  
    if (n > 100):  
        return n - 10  
    return fun(fun(n + 11))  
print(fun(96))
```

Put Your Solution Here:

9. Write a Program to find the length of a string using recursion. Ask for the user input to test the code.

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10. Write a recursive function named zigzag that returns a string of n characters as follows. The middle character (or the middle two characters if n is even) is an asterisk (*). All characters before the asterisks are '<'. All characters after are '>'. Report an error if n is not positive. Ask for the user input to test the code.

Example: n =6, pattern is <<*>>

if n=5 pattern is <<*>

Put Your Solution Here:

11. In a sorted array containing prime numbers from 2 to 311, approximately how many elements would a binary search algorithm need to examine before determining that 52 is not present in the array, indicating that 52 is not a prime number?

Put Your Solution Here:

Explanation:

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12. What is the Big O notation for the time complexity of the calculate_power function in terms of the input parameters base_number and exp? Explain? You can assume that multiplying two numbers takes time $O(1)$.

```
def calculate_power(base_number, exp):  
    if exp == 0:  
        return 1  
    elif exp % 2 == 0:  
        power = calculate_power(base_number, exp // 2)  
        return power * power  
    else:  
        power = calculate_power(base_number, exp // 2)  
        return base_number * power * power
```

Put Your Solution Here:

Explanation:

-
13. Let's consider a recursive function g, where g(0) is defined as 1 and g(n) is defined as $2 * g(n-1) + n$. Calculate the value of g(5) by writing the step-by-step process.

Put Your Solution Here:

14. How many times does the below function recur? Write the recursive trace of this program

```
def func(x):  
    if(x>20):  
        return 0  
    if(x%2):  
        return 1+func(x+1)  
    return 1+func(x+3)  
print(func(1))
```

Put your Solution here:

F

15. Understand the code below, write the following code's output, and explain how this output is achieved.

```
LIMIT = 1000
def fun2(n):
    if (n <= 0):
        return
    if (n > LIMIT):
        return
    print(n, end=" ")
    fun2(2 * n)
    print(n, end=" ")

print(fun2(7))
```

Put your solution here:

Explanation:

F